

Лабораторная работа по Git

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1) Клонирование репозитория

```
MacBook-Air-5:FirstLab kaplanianarman$ git clone https://github.com/smartiqaorg/geometric_lib
Cloning into 'geometric_lib'...
remote: Enumerating objects: 35, done.
remote: Total 35 (delta 0), reused 0 (delta 0), pack-reused 35 (from 1)
Receiving objects: 100% (35/35), 4.80 KiB | 2.40 MiB/s, done.
Resolving deltas: 100% (4/4), done.
MacBook-Air-5:FirstLab kaplanianarman$ pwd
/Users/kaplanianarman/Desktop/FirstLab
MacBook-Air-5:FirstLab kaplanianarman$ cd geometric_lib/
MacBook-Air-5:geometric_lib kaplanianarman$
```

2) Создание новой ветки new_features532471

```
MacBook-Air-5:geometric_lib kaplanianarman$ git switch -c new_features532471
Switched to a new branch 'new_features532471'
MacBook-Air-5:geometric_lib kaplanianarman$
```

3) Добавление файла rectangle.py в ветку new_features532471, а также создание коммита с сообщением о добавлении нового файла rectangle.py

```
MacBook-Air-5:geometric_lib kaplanianarman$ git branch
main
* new_features532471
MacBook-Air-5:geometric_lib kaplanianarman$ pwd
/Users/kaplanianarman/Desktop/FirstLab/geometric_lib
MacBook-Air-5:geometric_lib kaplanianarman$ touch rectangle.py
MacBook-Air-5:geometric_lib kaplanianarman$ open rectangle.py
MacBook-Air-5:geometric_lib kaplanianarman$ git add .
MacBook-Air-5:geometric_lib kaplanianarman$ git commit -m "rectangle commit"
[new_features532471 94b7eb0] rectangle commit
1 file changed, 5 insertions(+)
create mode 100644 rectangle.py
MacBook-Air-5:geometric_lib kaplanianarman$
```

4) Добавление файла triangle.py в ветку new_features532471, исправление ошибки в файле rectangle.py, создание коммита с сообщением о внесенных изменениях

```
MacBook-Air-5:geometric_lib kaplanianarman$ ls
circle.py  docs  rectangle.py  square.py
MacBook-Air-5:geometric_lib kaplanianarman$ touch triangle.py
MacBook-Air-5:geometric_lib kaplanianarman$ open triangle.py
MacBook-Air-5:geometric_lib kaplanianarman$ open rectangle.py
MacBook-Air-5:geometric_lib kaplanianarman$ git add .
MacBook-Air-5:geometric_lib kaplanianarman$ git commit -m "adding triangle and fixing rectangle commit"
[new_features532471 0162f3e] adding triangle and fixing rectangle commit
2 files changed, 6 insertions(+), 1 deletion(-)
create mode 100644 triangle.py
MacBook-Air-5:geometric_lib kaplanianarman$
```

5) Построение графа истории всего репозитория с однострочным выводом коммитов

```

[MacBook-Air-5:geometric_lib kaplanianarman$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
[MacBook-Air-5:geometric_lib kaplanianarman$ git log --graph --oneline --all --decorate
* 0162f3e (new_features532471) adding triangle and fixing rectangle commit
* 94b7eb0 rectangle commit
| * 86edb1c (origin/release) L-05: Update Docs. Add user agreement info
| * 438b89a L-05: Add user agreement
| * 6adb962 L-03: Docs added
| | * 3049431 (origin/feature) L-04: Add rectangle.py
| | /
| | /
| | * b5b0fae (origin/develop) L-04: Update docs for calculate.py
| | * d76db2a L-04: Add calculate.py
| | * 51c40eb L-04: Doc updated for triangle
| | * d080c78 L-04: Triangle added
| | /
| | /
* | d078c8d (HEAD -> main, origin/main, origin/HEAD) L-03: Docs added
| /
* 8ba9aeb L-03: Circle and square added
MacBook-Air-5:geometric_lib kaplanianarman$ █

```

- 6) Построение графа с историей только текущей ветки с однострочным выводом последних двух коммитов

```

[MacBook-Air-5:geometric_lib kaplanianarman$ git switch new_features532471
Switched to branch 'new_features532471'
[MacBook-Air-5:geometric_lib kaplanianarman$ git log --graph --oneline --decorate -2
* 0162f3e (HEAD -> new_features532471) adding triangle and fixing rectangle commit
* 94b7eb0 rectangle commit
MacBook-Air-5:geometric_lib kaplanianarman$ █

```

- 7) Просмотр последних коммитов

```

[MacBook-Air-5:geometric_lib kaplanianarman$ git show 0162f3e
commit 0162f3ebe407977f472d44cabea4234e05e15a04 (HEAD -> new_features532471)
Author: Arman <kaplanianarman@gmail.com>
Date:   Fri Sep 4 22:18:06 2026 +0300

    adding triangle and fixing rectangle commit

diff --git a/rectangle.py b/rectangle.py
index 6b7ed43..c5b2252 100644
--- a/rectangle.py
+++ b/rectangle.py
@@ -2,4 +2,4 @@ def area(a, b):
     return a * b

def perimeter(a, b):
-    return a + b
+    return (a + b) * 2
diff --git a/triangle.py b/triangle.py
new file mode 100644
index 0000000..8871838
--- /dev/null
+++ b/triangle.py
@@ -0,0 +1,5 @@
+def area(a, h):
+    return a * h / 2
+
+def perimeter(a, b, c):
+    return a + b + c
MacBook-Air-5:geometric_lib kaplanianarman$

```

```

[MacBook-Air-5:geometric_lib kaplanianarman$ git show 94b7eb0
commit 94b7eb09786083758bde70c7c4a3232317ef0c2b
Author: Arman <kaplanianarman@gmail.com>
Date:   Fri Sep 4 22:16:13 2026 +0300

    rectangle commit

diff --git a/rectangle.py b/rectangle.py
new file mode 100644
index 0000000..6b7ed43
--- /dev/null
+++ b/rectangle.py
@@ -0,0 +1,5 @@
+def area(a, b):
+    return a * b
+
+def perimeter(a, b):
+    return a + b
MacBook-Air-5:geometric_lib kaplanianarman$

```

8) Удаление ветки new_features532471

```
[MacBook-Air-5:geometric_lib kaplanianarman$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
[MacBook-Air-5:geometric_lib kaplanianarman$ git branch -D new_features532471
Deleted branch new_features532471 (was 0162f3e).
[MacBook-Air-5:geometric_lib kaplanianarman$ ls
circle.py      docs           square.py
MacBook-Air-5:geometric_lib kaplanianarman$
```